bosonic string theory whose mass (more precisely, whose mass squared) was negative—a so-called tachyon. Even before string theory, physicists had studied the possibility that our world might have tachyon particles, in addition to the familiar particles that all have positive masses, but their efforts showed that it is difficult if not impossible for such a theory to be logically sensible. Similarly, in the context of bosonic string theory, physicists tried all sorts of fancy footwork to make sense of the bizarre prediction of a tachyon vibrational pattern, but to no avail. These features made it increasingly clear that although it was an interesting theory, the bosonic string was missing something essential.

In 1971, Pierre Ramond of the University of Florida took up the challenge of modifying the bosonic string theory to include fermionic patterns of vibration. Through his work and subsequent results of Schwarz and André Neveu, a new version of string theory began to emerge. And much to everyone's surprise, the bosonic and the fermionic patterns of vibration of this new theory appeared to come in pairs. For each bosonic pattern there was a fermionic pattern, and vice versa. By 1977, insights of Ferdinando Gliozzi of the University of Turin, Scherk, and David Olive of Imperial College put this pairing into the proper light. The new string theory incorporated supersymmetry, and the observed pairing of bosonic and fermionic vibrational patterns reflected this highly symmetric character. Supersymmetric string theory—superstring theory; that is—had been born. Moreover, the work of Gliozzi, Scherk, and Olive had one other crucial result: They showed that the troublesome tachyon vibration of the bosonic string does not afflict the superstring. Slowly, the pieces of the string puzzle were falling into place.

Nevertheless, the major initial impact of the work of Ramond, and also of Neveu and Schwarz, was not actually in string theory. By 1973, the physicists Julian Wess and Bruno Zumino realized that supersymmetry—the new symmetry emerging from the reformulation of string theory—was applicable even to theories based on point particles. They rapidly made important strides toward incorporating supersymmetry into the framework of point-particle quantum field theory. And since, at the time, quantum field theory was the central rage of the mainstream particle-physics community—with string